Team Yellow: Tetris

ASE 420 – Team Project Progress

Team contributions, current progress, and next steps

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What we built

Current features in the Tetris game

- The game is organized into separate classes and game states instead of one large file.
- The game includes a title screen, gameplay, pause menu, and game-over screen.
- Gameplay features include movement, rotation, scoring, levels, difficulty, hold piece, ghost piece, and next-piece preview.
- The project includes automated tests for gameplay, scoring, and high-score storage.

Team progress

Current project status

- The main gameplay systems are implemented and connected to the running game.
- The hold piece, landing ghost, and next-piece preview have been added.
- Scoring, levels, leaderboard, and game-over results are working in the game.
- The team is now focusing on polishing features, testing how they work together, and fixing demo feedback.
- Remaining work includes the difficulty picker, settings options, sound effects, theme selection, and restart flow.

Eurydice Lunnemann: Movement and game structure

Individual contribution

- Built the game structure, including the game engine, state classes, board, piece, and UML.
- Implemented timed soft drop, hard drop, and held-key repeat for left and right movement.
- Added rotation with wall kicks that follow standard Tetris rotation rules.
- Added a lock delay so a landed piece waits before locking.
- Added 22 tests for input handling, soft drop, lock delay, and wall kicks.

Daniel Than: Gameplay features

Individual contribution

- Implemented the hold-piece feature so players can save a piece and use it later.
- Added a landing ghost to show where the current piece will land.
- Added a preview of the next piece.
- Reported 94 lines of implementation code and 141 lines of test code.
- Added and passed 6 tests for the new features.
- Reported all 8 planned features and both assigned requirements complete.

Maryan Abdi: Team leader and user interface

Individual contribution

- Led Team Yellow and coordinated the team's work and feature assignments.
- Worked on the Start Menu and the user interface for the game.
- Worked on connecting the menu and game screens so the player can move through the game.
- Coordinated the integration of team members' code and helped resolve conflicts between features.
- Helped keep the overall game experience and team implementation moving together.

Software design ideas

How the project is organized

- State pattern: the game changes between title, play, pause, game-over, and settings states.
- Programming to interfaces: scoring and high-score storage use interfaces.
- Separation of responsibilities: gameplay, scoring, levels, storage, and interface code are kept in separate parts.
- Named constants make scoring values, fall speeds, and difficulty settings easier to change.
- Automated tests help check features and catch problems during development.

Next steps

Remaining work for the team

- Polish the completed gameplay features.
- Test all features together and fix issues found during the team demo.
- Connect difficulty selection to the menu.
- Finish settings options, sound effects, theme and color selection, and restart flow.
- Complete documentation and update the project schedule based on feedback.

Retrospective

What went well and what needs attention

What went well

- The team built the main game structure and added working gameplay features.
- Features were tested as they were developed.
- Team members contributed different parts of the game and worked on integrating them.

What needs attention

- Continue checking that new changes work with the rest of the game.
- Keep communication and testing consistent as the remaining features are completed.

Thank you

Questions?